

1 Introduction

1.1 For the impatient

Writing good colorblind-safe documents can be a tricky task. It is a good starting point to realize that the default L^AT_EX colors like red or blue should be avoided. This is why for the impatient, simply loading this package without specifying any options (i.e. `\usepackage{colorblind}`) redefines the default colors to more sensible alternatives.¹

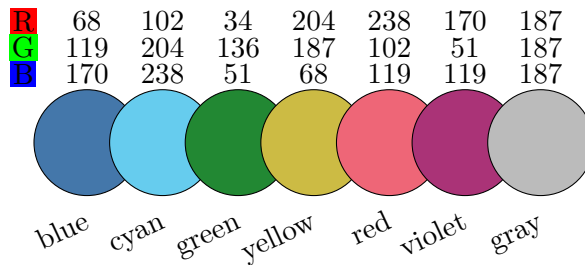


Figure 1: Redefinitions of the default L^AT_EX colors.

It should be noted that this is a very brute-force way of trying to achieve colorblind-safeness. If you care about this topic (which you should), the rest of this document provides details on how colorblind-safeness is best achieved in different scenarios.

1.2 Package options

Tol	The colorblind package provides the color schemes by Paul Tol [1] and the
OkabeIto	Okabe Ito color palette [2]. By default, no schemes are loaded. Providing one of the options Tol or OkabeIto loads all corresponding schemes.
pgf	If the option pgf is provided, continuous colormaps are defined for use with pgfplots (or TikZ). Also, the command <code>\drawSchemeC</code> for drawing continuous color schemes is only defined when the option is provided and continuous color schemes are available (through providing the Tol option). Continuous versions of color schemes are only available when the colors are allowed to be interpolated, see below for details.
no-tikz	The package uses TikZ to draw the discrete versions of color schemes. Providing the option no-tikz disables this, the command <code>\drawScheme</code> is not defined in this case.
keep-defaults	The package redefines the default colors like red or blue to be colorblind-safe. By specifying this option, the defaults are not changed.

¹Where no suitable redefinition was possible for a builtin named color, the definition is changed to black.